# Executive Overview & Architecture

Published: September 2026 · Reference Code: MAN-001

This architectural blueprint outlines the principles of local-first mobile software engineering. By confining all parsing, transformations, and indexing to the edge runtime, the application guarantees non-negotiable confidentiality, zero cloud latency, and resilience against network failures.

This architectural blueprint outlines the principles of local-first mobile software engineering. By confining all parsing, transformations, and indexing to the edge runtime, the application guarantees non-negotiable confidentiality, zero cloud latency, and resilience against network failures.

This architectural blueprint outlines the principles of local-first mobile software engineering. By confining all parsing, transformations, and indexing to the edge runtime, the application guarantees non-negotiable confidentiality, zero cloud latency, and resilience against network failures.

This architectural blueprint outlines the principles of local-first mobile software engineering. By confining all parsing, transformations, and indexing to the edge runtime, the application guarantees non-negotiable confidentiality, zero cloud latency, and resilience against network failures.

# Security Protocols & Cryptographic Keying

Published: September 2026 · Reference Code: MAN-002

Document security requires strict algorithmic isolation. Standard AES-256 with CBC/GCM modes and authenticated encryption ensures that files stored on flash storage remain immune to unauthorized inspection.

Document security requires strict algorithmic isolation. Standard AES-256 with CBC/GCM modes and authenticated encryption ensures that files stored on flash storage remain immune to unauthorized inspection.

Document security requires strict algorithmic isolation. Standard AES-256 with CBC/GCM modes and authenticated encryption ensures that files stored on flash storage remain immune to unauthorized inspection.

Document security requires strict algorithmic isolation. Standard AES-256 with CBC/GCM modes and authenticated encryption ensures that files stored on flash storage remain immune to unauthorized inspection.

# Memory Allocation & Native Pipeline Scaling

Published: September 2026 · Reference Code: MAN-003

Managing multi-gigabyte PDF streams within tight Android heap constraints (e.g. 192MB - 512MB limits) demands careful tile rendering, lazy bitmap recycling, and memory-mapped file handles.

Managing multi-gigabyte PDF streams within tight Android heap constraints (e.g. 192MB - 512MB limits) demands careful tile rendering, lazy bitmap recycling, and memory-mapped file handles.

Managing multi-gigabyte PDF streams within tight Android heap constraints (e.g. 192MB - 512MB limits) demands careful tile rendering, lazy bitmap recycling, and memory-mapped file handles.

Managing multi-gigabyte PDF streams within tight Android heap constraints (e.g. 192MB - 512MB limits) demands careful tile rendering, lazy bitmap recycling, and memory-mapped file handles.

# Typography, Fonts & Vector Layout Imposition

Published: September 2026 · Reference Code: MAN-004

Vector rendering fidelity depends on accurate font descriptor matching and CMap glyph translation. Imposition engines (2-Up, 4-Up, and Saddle-Stitch Booklets) calculate affine transformation matrices to map source media boxes onto target printer sheets without rasterization artifacts.

Vector rendering fidelity depends on accurate font descriptor matching and CMap glyph translation. Imposition engines (2-Up, 4-Up, and Saddle-Stitch Booklets) calculate affine transformation matrices to map source media boxes onto target printer sheets without rasterization artifacts.

Vector rendering fidelity depends on accurate font descriptor matching and CMap glyph translation. Imposition engines (2-Up, 4-Up, and Saddle-Stitch Booklets) calculate affine transformation matrices to map source media boxes onto target printer sheets without rasterization artifacts.

Vector rendering fidelity depends on accurate font descriptor matching and CMap glyph translation. Imposition engines (2-Up, 4-Up, and Saddle-Stitch Booklets) calculate affine transformation matrices to map source media boxes onto target printer sheets without rasterization artifacts.

# Form Flattening & Annotations Preservation

Published: September 2026 · Reference Code: MAN-005

AcroForm widgets often exhibit inconsistent visual states across diverse PDF viewers. By rasterizing or rendering appearance streams (/AP) directly into the page content stream (/Contents), signatures and checkboxes are locked permanently.

AcroForm widgets often exhibit inconsistent visual states across diverse PDF viewers. By rasterizing or rendering appearance streams (/AP) directly into the page content stream (/Contents), signatures and checkboxes are locked permanently.

AcroForm widgets often exhibit inconsistent visual states across diverse PDF viewers. By rasterizing or rendering appearance streams (/AP) directly into the page content stream (/Contents), signatures and checkboxes are locked permanently.

AcroForm widgets often exhibit inconsistent visual states across diverse PDF viewers. By rasterizing or rendering appearance streams (/AP) directly into the page content stream (/Contents), signatures and checkboxes are locked permanently.

# OCR Recognition & Spatial Layout Analysis

Published: September 2026 · Reference Code: MAN-006

Optical Character Recognition requires pre-processing pipelines: binarization, skew correction, and contrast normalization. Text bounding boxes are mapped to an invisible text layer over the source scan.

Optical Character Recognition requires pre-processing pipelines: binarization, skew correction, and contrast normalization. Text bounding boxes are mapped to an invisible text layer over the source scan.

Optical Character Recognition requires pre-processing pipelines: binarization, skew correction, and contrast normalization. Text bounding boxes are mapped to an invisible text layer over the source scan.

Optical Character Recognition requires pre-processing pipelines: binarization, skew correction, and contrast normalization. Text bounding boxes are mapped to an invisible text layer over the source scan.

# Document Repair & Damaged Header Recovery

Published: September 2026 · Reference Code: MAN-007

Truncated cross-reference tables (XREF) and corrupted trailer dictionaries represent 85% of PDF errors. The heuristic scanner reconstructs object offsets by traversing binary stream markers.

Truncated cross-reference tables (XREF) and corrupted trailer dictionaries represent 85% of PDF errors. The heuristic scanner reconstructs object offsets by traversing binary stream markers.

Truncated cross-reference tables (XREF) and corrupted trailer dictionaries represent 85% of PDF errors. The heuristic scanner reconstructs object offsets by traversing binary stream markers.

Truncated cross-reference tables (XREF) and corrupted trailer dictionaries represent 85% of PDF errors. The heuristic scanner reconstructs object offsets by traversing binary stream markers.

# Appendix: Compliance, Standards & Citations

Published: September 2026 · Reference Code: MAN-008

Conforms to ISO 32000-1, ISO 32000-2 (PDF 2.0), and PDF/A-1b archiving standards. All processing executed within Android sandbox with zero outbound network sockets.

Conforms to ISO 32000-1, ISO 32000-2 (PDF 2.0), and PDF/A-1b archiving standards. All processing executed within Android sandbox with zero outbound network sockets.

Conforms to ISO 32000-1, ISO 32000-2 (PDF 2.0), and PDF/A-1b archiving standards. All processing executed within Android sandbox with zero outbound network sockets.

Conforms to ISO 32000-1, ISO 32000-2 (PDF 2.0), and PDF/A-1b archiving standards. All processing executed within Android sandbox with zero outbound network sockets.